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**B.Sc. (E.C.S.) (Semester - III) (New) (NEP CBCS) Examination:
October/November – 2025
Statistics for Data Science – I (G06-DSC2-0303)**

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.
3) Use of log table and calculator is allowed.

06

- 1) Karl Pearson's coefficient of correlation lies between _____.
a) -1 to 0 b) 0 to 1
c) -1 to 1 d) None of these
- 2) Measures of central tendency that based on all observations is _____.
a) A.M. b) Median
c) Mode d) All of these
- 3) Line of regression X on Y is used to estimate _____.
a) X for known value of Y b) Y for known value of X
c) (A) and (B) both d) None of these
- 4) If all the observations are same in the given data then variance is _____.
a) Positive b) Negative
c) Zero d) None of these
- 5) Cumulative sum of frequencies from lowest class to highest class gives _____.
a) l.c.f. b) g.c.f.
c) relative frequency d) none of these
- 6) In _____ sampling method, same unit may be selected more than once.
a) SRS b) SRSWOR
c) SRSWR d) Stratified

06

- Define- class interval with example.
- Find first Quartile for the data: 14,18, 20,11,17, 21, 24, 10
- Define arithmetic mean.
- If $n=10$, $\Sigma x=14$, $\Sigma x^2=30$, find C.V.
- Find regression coefficient y on x if $r=0.5$, $\sigma_x=4$, $\sigma_y=10$.

Q.3 Answer the following questions. (Any two) **06**

- a) Explain systematic sampling method.
- b) Given: $n=10$, $\Sigma X=15$, $\Sigma Y=20$, $\Sigma XY=35$, $\Sigma X^2=55$, $\Sigma Y^2=88$ find 'r'.
- c) Explain scatter diagram method of studying correlation.

Q.4 Answer the following questions. (Any two) **06**

- a) Write a note on lines of regression.
- b) Find c.v. for the following frequency distribution.

X	10	12	14	16	18
f	4	7	11	9	2

- c) Find median for the following grouped data.

Class	5-10	10-15	15-20	20-25	25-30
f	11	13	17	12	9

Q.5 Answer the following question. (Any one) **06**

- a) Explain concept of measures of dispersion, state any two merits of S.D.
- b) Draw histogram to represent the following data and hence obtain mode.

Class	10-20	20-30	30-40	40-50	50-60	60-70
f	13	18	20	15	12	8